Exploiting Shared Physiological Signals from Both Ears for Heart Rate Estimation

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Abstract

Photoplethysmography (PPG) enables wearable heart-rate monitoring but is sensitive to motion artifacts. While ear-worn devices offer a more stable sensing site than the wrist, ear-based PPG remains affected by head and body movements. We propose a bilateral, motion-aware framework for heart-rate estimation using ear-worn PPG and IMU signals. Our method explicitly models the shared physiological component across the two ears while accounting for asymmetric corruption through motion-aware fusion. Evaluations on EarSet show that our approach consistently outperforms single-ear and naive bilateral fusion baselines.

CCS Concepts

• **Human-centered computing → Ubiquitous and mobile computing systems and tools**.

Keywords

Earables, Photoplethysmography, Heart Rate Monitoring

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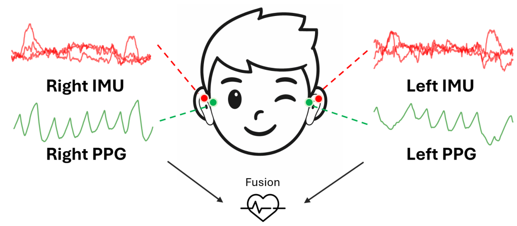

Figure 1: Illustration of Bilateral Ear Sensing.

1 Introduction

PPG has become a widely used modality for wearable physiological sensing, enabling continuous and non-invasive monitoring of signals such as heart rate. It is most commonly deployed on the wrist, but wrist-based PPG is highly vulnerable to motion artifacts, especially during daily activities and exercise, where large movements and unstable contact can severely degrade signal quality.

Ear-worn devices have recently emerged as a promising alternative for PPG sensing. Compared with the wrist, the head is relatively more stable, and the ear provides a compact and consistent sensing site. This has led to growing interest in in-ear physiological sensing in both research and emerging earable systems [1]. However, ear-based PPG still suffers from motion artifacts, not only from whole-body activities such as walking and running, but also from head and facial movements that perturb local contact conditions around the ear.

Prior work commonly uses IMU signals to guide motion artifact suppression, but such methods usually treat each sensing location independently. This overlooks a key property of earables: the two ears are driven by the wearer's underlying cardiac activity, even though the observed signals may be corrupted differently. As illustrated in Figure 1, bilateral ear signals can be viewed as correlated observations of a shared physiological process under different local noise conditions. This is particularly important for earables, where asymmetric facial or head movements may affect the two sides unevenly. Therefore, instead of suppressing noise from each ear in isolation, bilateral modeling offers an opportunity to reinforce the shared physiological component while reducing the influence of ear-specific corruption.

Motivated by this insight, we propose a bilateral, motion-aware framework for robust heart rate estimation using ear-worn PPG. Our method explicitly models the shared physiological component across the two ears while using IMU signals to guide motion-aware fusion under asymmetric corruption. Rather than reconstructing a clean waveform, the model directly learns a more robust cardiac representation for HR estimation by combining bilateral consistency with motion cues from both sides. Our evaluation on EarSet [3] shows that this simple bilateral design improves heart rate estimation accuracy over single-ear and naive fusion baselines.

2 System Design

As shown in Figure 2, our system decomposes each ear-worn PPG signal into a clean physiological feature and a motion artifact. It combines bilateral interaction with motion-aware fusion for robust heart-rate estimation. It consists of three modules: per-ear decomposition, bilateral interaction, and motion-aware fusion.

Per-ear decomposition. For each ear, the raw PPG and synchronized IMU signals are first encoded into latent features using shared encoders. The IMU feature captures motion dynamics, while the PPG feature contains both physiological and motion-corrupted

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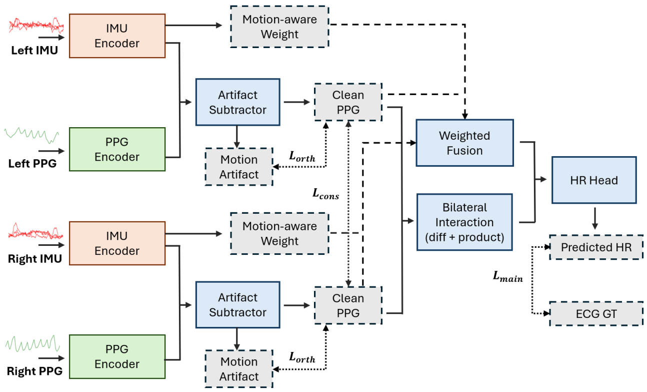

Figure 2: Overall architecture of the system.

information. We then estimate a motion artifact representation from the joint PPG–IMU features and subtract it from the PPG latent to obtain a clean physiological feature. This residual operation is implemented as a feature-level subtraction, enforcing that the IMU-conditioned artifact branch explains motion-related variations, while the remaining component captures heartbeat-related information. In addition, we explicitly regularize the clean and artifact features to be disentangled.

Bilateral interaction. Given the clean features from both ears, we construct a shared representation that models the common physiological source. Instead of directly merging the two features, we include symmetric interaction terms, including their difference and element-wise product, to capture both consistency and discrepancy across ears. This design emphasizes the component that is jointly supported by both sides while remaining aware of asymmetric corruption.

Motion-aware fusion. To handle asymmetric noise, we further introduce a motion-aware fusion mechanism. The IMU features from both ears are used to estimate reliability weights, which reflect the relative quality of each side under current motion. These weights are then used to compute a weighted combination of the two clean features, allowing the model to down-weight the more corrupted ear. The final heart rate is predicted from both the shared representation and the reliability-weighted fused representation.

Training objective. Let y be the ground-truth heart rate, $\hat{y}$ the predicted heart rate, and c_l, c_r and a_l, a_r denote the clean and artifact features for the left and right ears. We define the main regression loss as $\mathcal{L}_{\text{main}} = \text{Huber}(\hat{y}, y)$. To enforce the shared physiological source, we introduce a bilateral consistency loss, $\mathcal{L}_{\text{cons}} = 1 - \cos(c_l, c_r)$, which encourages the clean features from both ears to be aligned in the latent space. To disentangle physiological signals and motion artifacts, we impose an orthogonality constraint, $\mathcal{L}_{\text{orth}} = \frac{1}{2}\left(|\langle \tilde{c}_l, \tilde{a}_l \rangle| + |\langle \tilde{c}_r, \tilde{a}_r \rangle|\right)$, where $\tilde{c}$ and $\tilde{a}$ denote ℓ_2-normalized features. The total loss is defined as $\mathcal{L} = \mathcal{L}_{\text{main}} + 0.2\,\mathcal{L}_{\text{cons}} + 0.05\,\mathcal{L}_{\text{orth}}$, which enforces accurate heart-rate estimation, cross-ear consistency, and physiology-artifact disentanglement.

3 Performance Evaluation

We evaluate our system on the EarSet dataset [3]. We randomly select 10 subjects and split them into training, validation, and test sets with a ratio of 7:2:1 at the subject level to avoid subject leakage and ensure cross-subject evaluation. To focus on a practical sensing setup, we only use the green-light PPG channel and the 3-axis accelerometer signals. Both PPG and IMU are segmented using 8 s windows, and the average ECG-derived heart rate within each window is used as the ground truth.

We compare our method with two baselines. The first is a classical single-ear motion-aware heart-rate estimation baseline [2], which takes PPG and IMU from one ear as input and uses motion information to suppress artifacts. This type of method is widely adopted in prior wearable heart-rate estimation systems, but it processes each side independently and cannot exploit bilateral physiological consistency. The second is a dual-ear baseline, which directly fuses bilateral signals using a structure similar to the single-ear model, but without explicitly modeling cross-ear consistency or motion-aware reliability. This baseline is designed to examine how much improvement can be obtained by simply introducing bilateral data. Finally, we evaluate our full model to quantify the benefit of our explicit decomposition, bilateral interaction, and motion-aware fusion design.

Table 1: Performance (MAE in bpm) across activities.

Method	Stationary	One-shot Movements	Continuous Movements
Left Ear	6.43	8.72	8.01
Right Ear	5.92	7.73	7.34
Both Ears	4.74	7.18	7.04
Our System	2.86	3.46	4.37

Table 1 reports the results. As shown, single-ear methods achieve limited performance due to their sensitivity to motion artifacts, even under these relatively mild activities. One possible reason is that the same motion-aware design is used across both stationary and dynamic conditions, while IMU cues may be less informative in low-motion settings and may introduce unnecessary variability. Simply combining both ears provides only limited improvement and remains suboptimal under dynamic conditions, likely because motion corruption is often asymmetric across the two sides and naive fusion cannot distinguish the more reliable ear from the more corrupted one. In contrast, our system consistently outperforms all baselines across all activity types, reducing the MAE from 4.74 to 2.86 under stationary conditions, from 5.18 to 3.46 under one-shot movements, and from 6.04 to 4.37 under continuous movements. These results demonstrate that explicitly modeling bilateral consistency and motion-aware fusion is effective for robust heart-rate estimation.

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